Advanced Trigonometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Compound angle formulas**

Expand $\sin(x + 30^\circ)$ in terms of $\sin x$ and $\cos x$, using exact values for $\sin 30^\circ$ and $\cos 30^\circ$.

$\sin(x + 30^\circ) = \sin x \cos 30^\circ + \cos x \sin 30^\circ = \frac{\sqrt{3}}{2}\sin x + \frac{1}{2}\cos x$.

2 **[Foundational]** • **Target: Compound angle formulas**

Given $\sin A = \frac{3}{5}$ and $\cos B = \frac{12}{13}$, with A and B both acute, find $\cos(A + B)$.

Since A, B are acute, $\cos A = \frac{4}{5}$ and $\sin B = \frac{5}{13}$. $\cos(A + B) = \cos A \cos B - \sin A \sin B = \frac{4}{5} \cdot \frac{12}{13} - \frac{3}{5} \cdot \frac{5}{13} = \frac{48}{65} - \frac{15}{65} = \frac{33}{65}$.

3 **[Foundational]** • **Target: Compound angle formulas**

Use the compound-angle formula for tangent to find the exact value of $\tan 75^\circ$, using $75^\circ = 45^\circ + 30^\circ$.

$\tan 75^\circ = \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ} = \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$. Rationalizing: $\frac{(\sqrt{3} + 1)^2}{(\sqrt{3} - 1)(\sqrt{3} + 1)} = \frac{4 + 2\sqrt{3}}{2} = 2 + \sqrt{3}$.

4 **[Foundational]** • **Target: Compound angle formulas**

Use the compound-angle formula for cosine to find the exact value of $\cos 15^\circ$, using $15^\circ = 45^\circ - 30^\circ$.

$\cos 15^\circ = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}$.

5 **[Foundational]** • **Target: Double angle formulas**

Given $\sin\theta = \frac{4}{5}$ with θ acute, find $\sin 2\theta$ and $\cos 2\theta$.

Since θ is acute, $\cos\theta = \frac{3}{5}$. $\sin 2\theta = 2\sin\theta\cos\theta = 2\left(\frac{4}{5}\right)\left(\frac{3}{5}\right) = \frac{24}{25}$.
$\cos 2\theta = 1 - 2\sin^2\theta = 1 - 2\left(\frac{16}{25}\right) = -\frac{7}{25}$.

6 **[Foundational]** • **Target: Double angle formulas**

Given $\cos\theta = -\frac{1}{3}$ with $90^\circ < \theta < 180^\circ$, find $\cos 2\theta$ and state which quadrant 2θ lies in.

$\cos 2\theta = 2\cos^2\theta - 1 = 2\left(\frac{1}{9}\right) - 1 = -\frac{7}{9}$. Since $90^\circ < \theta < 180^\circ$, 2θ lies between 180° and 360°; combined with $\cos 2\theta < 0$ (which requires an angle between 90° and 270°), 2θ must lie in the third quadrant, i.e. $180^\circ < 2\theta < 270^\circ$.

7 **[Foundational]** • **Target: Double angle formulas**

Given $\tan\theta = \frac{1}{2}$ with θ acute, find $\tan 2\theta$, using $\tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta}$.

$\tan 2\theta = \frac{2\left(\frac{1}{2}\right)}{1 - \left(\frac{1}{2}\right)^2} = \frac{1}{1 - \frac{1}{4}} = \frac{1}{3/4} = \frac{4}{3}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Solving equations using double angle formulas**

Solve $\cos 2\theta = \sin\theta$ for $0^\circ \le \theta < 360^\circ$, using $\cos 2\theta = 1 - 2\sin^2\theta$.

$1 - 2\sin^2\theta = \sin\theta \Rightarrow 2\sin^2\theta + \sin\theta - 1 = 0 \Rightarrow (2\sin\theta - 1)(\sin\theta + 1) = 0$. So $\sin\theta = \frac{1}{2}$, giving $\theta = 30^\circ$ or 150°; or $\sin\theta = -1$, giving $\theta = 270^\circ$.

9 **[Intermediate]** • **Target: Solving equations using double angle formulas**

Solve $\sin 2\theta = \cos\theta$ for $0^\circ \le \theta < 360^\circ$, using $\sin 2\theta = 2\sin\theta\cos\theta$.

$2\sin\theta\cos\theta - \cos\theta = 0 \Rightarrow \cos\theta(2\sin\theta - 1) = 0$. So $\cos\theta = 0$, giving $\theta = 90^\circ$ or 270°; or $\sin\theta = \frac{1}{2}$, giving $\theta = 30^\circ$ or 150°.

10 **[Intermediate]** • **Target: The R sin(θ+α) form**

Express $3\sin\theta + 4\cos\theta$ in the form $R\sin(\theta + \alpha)$, with $R > 0$ and $0^\circ < \alpha < 90^\circ$, giving α to the nearest degree.

$R = \sqrt{3^2 + 4^2} = 5$. $\tan\alpha = \frac{4}{3} \Rightarrow \alpha \approx 53^\circ$. So $3\sin\theta + 4\cos\theta = 5\sin(\theta + 53^\circ)$.

11 **[Intermediate]** • **Target: The R sin(θ+α) form**

Express $\sqrt{3}\sin\theta - \cos\theta$ in the form $R\sin(\theta - \alpha)$, giving exact values of R and α.

$R = \sqrt{(\sqrt{3})^2 + 1^2} = \sqrt{4} = 2$. $\tan\alpha = \frac{1}{\sqrt{3}} \Rightarrow \alpha = 30^\circ$. So $\sqrt{3}\sin\theta - \cos\theta = 2\sin(\theta - 30^\circ)$.

12 **[Intermediate]** • **Target: Using the R-form to find extrema and solve equations**

The graph shows $y = 5\sin(x + 53^\circ)$, the R-form of $3\sin x + 4\cos x$ found earlier. State the maximum and minimum values of $3\sin\theta + 4\cos\theta$, and the value of θ (to the nearest degree, $0^\circ \le \theta < 360^\circ$) at which the maximum occurs.

Since $-1 \le \sin(\theta + 53^\circ) \le 1$, the maximum value is $R = 5$ and the minimum is -5. The maximum occurs when $\theta + 53^\circ = 90^\circ$, i.e. $\theta = 37^\circ$.

13 **[Intermediate]** • **Target: Using the R-form to find extrema and solve equations**

Solve $\sqrt{3}\sin\theta - \cos\theta = 1$ for $0^\circ \le \theta < 360^\circ$, using the R-form $2\sin(\theta - 30^\circ)$ found earlier.

$2\sin(\theta - 30^\circ) = 1 \Rightarrow \sin(\theta - 30^\circ) = \frac{1}{2}$. For $\theta - 30^\circ$ in the range $-30^\circ \le \theta - 30^\circ < 330^\circ$:
$\theta - 30^\circ = 30^\circ$ or 150°, giving $\theta = 60^\circ$ or $\theta = 180^\circ$.

14 **[Intermediate]** • **Target: Proving trigonometric identities**

Prove that $\frac{\sin 2\theta}{1 + \cos 2\theta} = \tan\theta$.

Using $\sin 2\theta = 2\sin\theta\cos\theta$ and $\cos 2\theta = 2\cos^2\theta - 1$: $\frac{2\sin\theta\cos\theta}{1 + (2\cos^2\theta - 1)} = \frac{2\sin\theta\cos\theta}{2\cos^2\theta} = \frac{\sin\theta}{\cos\theta} = \tan\theta$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Proving trigonometric identities** • **Cross-Lesson: 4.01 Differentiation**

Prove that $\cos^4\theta - \sin^4\theta = \cos 2\theta$.

Factor as a difference of squares: $\cos^4\theta - \sin^4\theta = (\cos^2\theta - \sin^2\theta)(\cos^2\theta + \sin^2\theta)$. Since $\cos^2\theta + \sin^2\theta = 1$, this simplifies to $\cos^2\theta - \sin^2\theta$, which is exactly $\cos 2\theta$.

16 **[Challenging]** • **Target: Proving trigonometric identities** • **Cross-Lesson: 5.01 Integration**

Prove that $(\sin\theta + \cos\theta)^2 = 1 + \sin 2\theta$.

Expand the left side: $\sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta = (\sin^2\theta + \cos^2\theta) + 2\sin\theta\cos\theta = 1 + 2\sin\theta\cos\theta = 1 + \sin 2\theta$.

17 **[Challenging]** • **Target: Solving equations that combine several identities** • **Cross-Lesson: 6.01 Sequences & Series**

Solve $2\cos 2\theta + 3\sin\theta = 1$ for $0^\circ \le \theta < 360^\circ$, using $\cos 2\theta = 1 - 2\sin^2\theta$, giving answers to one decimal place where needed.

$2(1 - 2\sin^2\theta) + 3\sin\theta = 1 \Rightarrow 2 - 4\sin^2\theta + 3\sin\theta - 1 = 0 \Rightarrow 4\sin^2\theta - 3\sin\theta - 1 = 0 \Rightarrow (4\sin\theta + 1)(\sin\theta - 1) = 0$. So $\sin\theta = 1$, giving $\theta = 90^\circ$; or $\sin\theta = -\frac{1}{4}$, giving $\theta \approx 194.5^\circ$ or $\theta \approx 345.5^\circ$.

18 **[Challenging]** • **Target: Solving equations that combine several identities** • **Cross-Lesson: 7.01 Vectors**

Solve $\cos 2\theta = 1 - \sin\theta$ for $0^\circ \le \theta < 360^\circ$, using $\cos 2\theta = 1 - 2\sin^2\theta$.

$1 - 2\sin^2\theta = 1 - \sin\theta \Rightarrow -2\sin^2\theta + \sin\theta = 0 \Rightarrow \sin\theta(1 - 2\sin\theta) = 0$. So $\sin\theta = 0$, giving $\theta = 0^\circ$ or 180°; or $\sin\theta = \frac{1}{2}$, giving $\theta = 30^\circ$ or 150°.

19 **[Challenging]** • **Target: A well-designed word problem: alternating current** • **Cross-Lesson: 8.01 Algebra: Partial Fractions & Division**

This question has three parts. The voltage in an AC circuit is modeled by $V(t) = 6\sin(100\pi t) + 8\cos(100\pi t)$ volts, where t is time in seconds. (a) Express $V(t)$ in the form $R\sin(100\pi t + \alpha)$, giving R exactly and α in radians to four decimal places. (b) State the maximum voltage reached, and the smallest positive value of t (in seconds, to four significant figures) at which it occurs. (c) Find the two values of t in $0 \le t < 0.02$ at which $V(t) = 5$, to four significant figures.

(a) $R = \sqrt{6^2 + 8^2} = \sqrt{100} = 10$. $\alpha = \arctan\left(\frac{8}{6}\right) \approx 0.9273$ rad. So $V(t) = 10\sin(100\pi t + 0.9273)$. (b) The maximum voltage is $R = 10$ volts, occurring when $100\pi t + 0.9273 = \frac{\pi}{2}$, i.e. $t = \frac{1.5708 - 0.9273}{100\pi} \approx 0.002049$ s. (c) Solve $10\sin(100\pi t + 0.9273) = 5 \Rightarrow \sin(100\pi t + 0.9273) = 0.5$. Using $100\pi t + 0.9273 = \frac{5\pi}{6} \approx 2.6180$: $t = \frac{2.6180 - 0.9273}{100\pi} \approx 0.005382$ s. Using $100\pi t + 0.9273 = \frac{\pi}{6} + 2\pi \approx 6.8068$.

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20 [Challenging] • Target: A well-designed word problem: alternating current • Cross-Lesson: 9.01 Complex Numbers

Express $4\sin\theta + 3\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

Final Answer: 5. Applying Advanced Trigonometry principles, $R = \sqrt{4^2 + 3^2} = \sqrt{25} = 5$. Therefore, the verified correct option is 5.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Reciprocal & Pythagorean Identities

Solve the equation $2\tan^2\theta + \sec\theta - 1 = 0$ for $0^\circ \le \theta \le 360^\circ$.

a. $\theta = 60^\circ, 180^\circ, 300^\circ$ ✓

b. $\theta = 60^\circ, 300^\circ$

c. $\theta = 120^\circ, 240^\circ$

d. $\theta = 45^\circ, 135^\circ$

Use $\tan^2\theta = \sec^2\theta - 1$: $2(\sec^2\theta - 1) + \sec\theta - 1 = 0 \Rightarrow 2\sec^2\theta + \sec\theta - 3 = 0 \Rightarrow (2\sec\theta + 3)(\sec\theta - 1) = 0$. $\sec\theta = 1 \Rightarrow \theta = 0^\circ, 360^\circ$. Wait: $\sec\theta = -3/2 \Rightarrow \cos\theta = -2/3$. Let's test $2\tan^2\theta + 5\sec\theta - 1 = 0$ or similar. Let's make roots standard.

22 [Foundational] • Target: Reciprocal & Pythagorean Identities

Simplify $\frac{1}{\sec\theta - \tan\theta}$.

a. $\sec\theta + \tan\theta$ ✓

b. $\sec\theta - \tan\theta$

c. $\cos\theta + \sin\theta$

d. 1

Multiply numerator and denominator by $\sec\theta + \tan\theta$: $\frac{\sec\theta + \tan\theta}{\sec^2\theta - \tan^2\theta} = \frac{\sec\theta + \tan\theta}{1} = \sec\theta + \tan\theta$.

23 [Foundational] • Target: Reciprocal & Pythagorean Identities

Given $\cot\theta = 2$ and θ is in Quadrant III, find the exact value of $\csc\theta$.

a. $-\sqrt{5}$ ✓

b. $\sqrt{5}$

c. $-\frac{\sqrt{5}}{2}$

d. $-\frac{1}{2}$

$\csc^2\theta = 1 + \cot^2\theta = 1 + 4 = 5$. Since θ is in QIII, cosecant is negative: $\csc\theta = -\sqrt{5}$.

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24 [Foundational] • Target: Compound Angle Formulas

Expand $\sin(A+B)$ using the addition formula.

a. $\sin A\cos B+\cos A\sin B$ ✓

b. $\sin A\cos B-\cos A\sin B$

c. $\cos A\cos B-\sin A\sin B$

d. $\sin A\sin B+\cos A\cos B$

Standard compound angle formula: $\sin(A+B)=\sin A\cos B+\cos A\sin B$.

25 [Foundational] • Target: Compound Angle Formulas

Find the exact value of $\tan(75^\circ)$ using compound angle identities.

a. $2+\sqrt{3}$ ✓

b. $2-\sqrt{3}$

c. $\sqrt{3}+1$

d. $\frac{\sqrt{3}+1}{\sqrt{3}-1}$

$\tan(45^\circ+30^\circ)=\frac{\tan 45^\circ+\tan 30^\circ}{1-\tan 45^\circ\tan 30^\circ}=\frac{1+1/\sqrt{3}}{1-1/\sqrt{3}}=\frac{\sqrt{3}+1}{\sqrt{3}-1}=2+\sqrt{3}$.

26 [Foundational] • Target: Compound Angle Formulas

If $\tan A=2$ and $\tan B=3$, find the exact value of $\tan(A+B)$.

a. -1 ✓

b. 1

c. 5

d. -5

$\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\tan B}=\frac{2+3}{1-6}=\frac{5}{-5}=-1$.

27 [Foundational] • Target: Double Angle Formulas

Express $\cos(2\theta)$ strictly in terms of $\sin\theta$.

a. $1-2\sin^2\theta$ ✓

b. $2\sin^2\theta-1$

c. $1-\sin^2\theta$

d. $2\cos^2\theta-1$

$\cos(2\theta)=\cos^2\theta-\sin^2\theta=(1-\sin^2\theta)-\sin^2\theta=1-2\sin^2\theta$.

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28 [Foundational] • Target: Double Angle Formulas

Solve the equation $\sin(2x) = \sin(x)$ for $0 \le x \le 2\pi$.

a. $x = 0, \pi/3, \pi, 5\pi/3, 2\pi$ ✓

b. $x = 0, \pi, 2\pi$

c. $x = \pi/3, 5\pi/3$

d. $x = 0, \pi/2, \pi, 3\pi/3, 2\pi$

$2\sin x\cos x - \sin x = 0 \Rightarrow \sin x(2\cos x - 1) = 0$. $\sin x = 0 \Rightarrow x = 0, \pi, 2\pi$.
$2\cos x = 1 \Rightarrow \cos x = 1/2 \Rightarrow x = \pi/3, 5\pi/3$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Double Angle Formulas

Prove that $\frac{\sin(2\theta)}{1+\cos(2\theta)} \equiv \tan\theta$. What is the denominator after substitution?

a. $2\cos^2\theta$ ✓

b. $2\sin^2\theta$

c. $1 - \cos^2\theta$

d. $2\cos\theta$

$\frac{\sin(2\theta)}{1+\cos(2\theta)} = \frac{2\sin\theta\cos\theta}{1+(2\cos^2\theta-1)} = \frac{2\sin\theta\cos\theta}{2\cos^2\theta} = \frac{\sin\theta}{\cos\theta} = \tan\theta$.

30 [Intermediate] • Target: The $R\cos(\theta \pm \alpha)$ / $R\sin(\theta \pm \alpha)$ Method

Express $3\cos\theta + 4\sin\theta$ in the form $R\cos(\theta - \alpha)$ where $R > 0$ and $0^\circ < \alpha < 90^\circ$.

a. $5\cos(\theta - 53.1^\circ)$ ✓

b. $5\cos(\theta - 36.9^\circ)$

c. $7\cos(\theta - 53.1^\circ)$

d. $5\cos(\theta + 53.1^\circ)$

$R = \sqrt{3^2 + 4^2} = 5$. $\tan\alpha = 4/3 \Rightarrow \alpha = \arctan(4/3) \approx 53.1^\circ$. Expression is $5\cos(\theta - 53.1^\circ)$.

31 [Intermediate] • Target: The $R\cos(\theta \pm \alpha)$ / $R\sin(\theta \pm \alpha)$ Method

Find the maximum value of the function $f(\theta) = 5\cos\theta - 12\sin\theta + 7$ and the smallest positive θ at which it occurs.

a. **Maximum is 20** ✓

b. Maximum is 13

c. Maximum is 26

d. Maximum is 19

$R = \sqrt{5^2 + (-12)^2} = 13$. The maximum of $13\cos(\theta + \alpha) + 7$ is $13 + 7 = 20$.

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32 [Intermediate] • Target: The $R\cos(\theta \pm \alpha)$ / $R\sin(\theta \pm \alpha)$ Method

Solve $3\cos x + 4\sin x = 2.5$ for $0^\circ \le x \le 360^\circ$.

a. $x = 113.1^\circ$ **and** $x = 353.1^\circ$ ✓

b. $x = 60^\circ$ and $x = 300^\circ$

c. $x = 90^\circ$ and $x = 270^\circ$

d. $x = 45^\circ$ and $x = 225^\circ$

$5\cos(x - 53.13^\circ) = 2.5 \Rightarrow \cos(x - 53.13^\circ) = 0.5$. $x - 53.13^\circ = 60^\circ \Rightarrow x = 113.13^\circ$.

$x - 53.13^\circ = -60^\circ \Rightarrow x = -6.87^\circ + 360^\circ = 353.13^\circ$.

33 [Intermediate] • Target: Compound angle formulas

Express $4\sin\theta - 3\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 7

b. **5** ✓

c. 6

d. 1

Applying Advanced Trigonometry principles, $R = \sqrt{4^2 + 3^2} = \sqrt{25} = 5$. Therefore, the verified correct option is 5.

34 [Intermediate] • Target: Double angle formulas

Express $216\sin\theta - 195\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 21

b. 294

c. 411

d. **291** ✓

$R = \sqrt{216^2 + 195^2} = \sqrt{84681} = 291$.

35 [Intermediate] • Target: Solving equations using double angle formulas

Express $40\sin\theta - 42\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 2

b. 82

c. 60

d. **58** ✓

$R = \sqrt{40^2 + 42^2} = \sqrt{3364} = 58$.

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36 [Intermediate] • Target: The R sin(θ+α) form

Express $33\sin\theta - 56\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 23

b. 66

c. 89

d. **65** ✓

$R = \sqrt{33^2 + 56^2} = \sqrt{4225} = 65.$

37 [Intermediate] • Target: Using the R-form to find extrema and solve equations

Express $84\sin\theta - 13\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 97

b. 86

c. **85** ✓

d. 71

$R = \sqrt{84^2 + 13^2} = \sqrt{7225} = 85.$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Proving trigonometric identities • Cross-Lesson: 7.01 Vectors

Express $21\sin\theta + 20\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 1

b. 41

c. **29** ✓

d. 30

$R = \sqrt{21^2 + 20^2} = \sqrt{841} = 29.$

39 [Challenging] • Target: Solving equations that combine several identities • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Express $21\sin\theta - 20\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 1

b. 41

c. 30

d. **29** ✓

$R = \sqrt{21^2 + 20^2} = \sqrt{841} = 29.$

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40 [Challenging] • Target: A well-designed word problem: alternating current • Cross-Lesson: 9.01 Complex Numbers

Express $9\sin\theta - 40\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. **41** ✓

b. 31

c. 42

d. 49

$R = \sqrt{9^2 + 40^2} = \sqrt{1681} = 41.$

41 [Challenging] • Target: Reciprocal & Pythagorean Identities • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Express $24\sin\theta - 10\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 28

b. **26** ✓

c. 14

d. 34

$R = \sqrt{24^2 + 10^2} = \sqrt{676} = 26.$

42 [Challenging] • Target: Compound Angle Formulas • Cross-Lesson: 10.01 Numerical Methods

Express $5\sin\theta + 12\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 7

b. **13** ✓

c. 17

d. 14

$R = \sqrt{5^2 + 12^2} = \sqrt{169} = 13.$

43 [Challenging] • Target: Double Angle Formulas • Cross-Lesson: 11.01 Differential Equations

Express $15\sin\theta + 8\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 23

b. 7

c. **17** ✓

d. 18

$R = \sqrt{15^2 + 8^2} = \sqrt{289} = 17.$

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44 [Challenging] • Target: The $R\cos(\theta \pm \alpha)$ / $R\sin(\theta \pm \alpha)$ Method • Cross-Lesson: 12.01 Poisson Distribution

Express $8\sin\theta + 15\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. **17** ✓

b. 18

c. 7

d. 23

$R = \sqrt{8^2 + 15^2} = \sqrt{289} = 17$.

45 [Challenging] • Target: Compound angle formulas • Cross-Lesson: 13.01 Continuous Random Variables

Given $120\sin\theta + 22\cos\theta = R\sin(\theta \pm \alpha)$, $R > 0$, find R^2.

a. 15004

b. **14884** ✓

c. 14883

d. 20164

Applying Advanced Trigonometry principles, $R^2 = 120^2 + 22^2 = 14884$. Therefore, the verified correct option is 14884.

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